pyAmpliCol at one loop

The recursive one-loop engine against MadLoop, Recola, OpenLoops and NJet

2026-10-24 · engine `b4d76cc3` on `loop`, timed at `1c971289` · AMD EPYC 9754, shared host · binary64 · one pinned thread · batches of $B = 128$

Reading guide. One row per test process, by multiplicity. A cell gives pyAmpliCol's value in bold and, at its right, the multiplier *pyAmpliCol ÷ the fastest provider* on this host (in bold among the provider values); the colour encodes it (blue faster, red slower). Main values are the engine on `loop` (gate-G1 campaign, 2026-10-24); lines tagged DEV are the WP3.1 development configuration with native per-point inputs and never set a colour. Grey cells say why a value is missing. Run times: per phase-space point in batches, finite part, stability checks off; providers in production mode (Section 1).

Table 1: **State of the one-loop engine.** Every value from `data/summary.json` (`scripts/build_summary.py`).

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at the largest batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
2 → 2	$gg \to gg$ 4 legs · QCD · 24 families OL·Rc·ML ≤ 8.7e−13 16 pts · median 2.2e−14	**10.4 s** ×73 0.34 GiB peak · plan 0.321 MB **Rc 142 ms** · ML 9.70 s DEV native 9.82 s ×69	**132 ms** ×66 **ML 2.02 ms** · OL 3.20 ms · Rc 10.0 ms $B = 128$ no gate (2→2) inputs 97 % DEV native 2.22 ms ×1.1	*not timed on loop*
2 → 3	$u\bar{u} \to d\bar{d}g$ 5 legs · QCD · 42 families (LC 27) ML·OL ≤ 9.2e−13 · Rc 3.3e−10 16 pts · median 8.1e−14	**9.94 s** ×24 0.34 GiB peak · plan 0.300 MB **Rc 406 ms** · ML 9.87 s DEV native 10.6 s ×26	**32.4 ms** ×21 **OL 1.52 ms** · ML 3.17 ms · Rc 3.18 ms $B = 128$ fail ×16 inputs 87 % DEV native 2.65 ms ×1.8 fail ×1.3	**37.4 ms** LC/FC 1.15 $B = 128$ · FC 32.4 ms DEV native 1.91 ms · LC/FC 0.72
	$u\bar{u} \to u\bar{u}g$ 5 legs · QCD · 68 families (LC 42) OL·ML ≤ 1.3e−11 · Rc 7.2e−11 16 pts · median 1.3e−13	**10.4 s** ×23 0.34 GiB peak · plan 0.424 MB **Rc 462 ms** · ML 10.9 s DEV native 10.9 s ×24	**54.5 ms** ×19 **OL 2.92 ms** · Rc 5.55 ms · ML 8.59 ms $B = 128$ fail ×14 inputs 89 % DEV native 3.80 ms ×1.3 pass‡ ×1.0	*not timed on loop* DEV native 3.16 ms · LC/FC 0.83
	$u\bar{u} \to e^-e^+g$ 5 legs · QCD · 11 families Rc·ML ≤ 8.8e−14 DEV 16 pts · median 8.3e−15 · †	*refused on loop* **Rc 701 ms** · ML 14.0 s DEV native 26.0 s ×37 · 0.35 GiB · 0.310 MB	*refused on loop: Z fermion signs (WP3.2), Furry zeros (WP3.5)* **ML 0.948 ms** · Rc 1.08 ms · OL no library DEV native 0.772 ms ×0.82 pass ×0.82 DEV numerator 34 %	*refused on loop*
	$u\bar{u} \to ggg$ 5 legs · QCD · 90 families (LC 62) ML·Rc ≤ 1.5e−12 · OL 8.4e−11 16 pts · median 5.4e−14	**12.3 s** ×19 0.36 GiB peak · plan 0.777 MB **Rc 634 ms** · ML 17.6 s DEV native 12.2 s ×19	**179 ms** ×14 **OL 12.4 ms** · Rc 25.3 ms · ML 28.2 ms $B = 128$ fail ×11 inputs 94 % DEV native 11.2 ms ×0.90 pass ×0.69	**171 ms** LC/FC 0.96 $B = 128$ · FC 179 ms DEV native 12.6 ms · LC/FC 1.13
	$gg \to ggg$ 5 legs · QCD · 154 families (LC 77) ML·Rc·OL ≤ 6.2e−7 16 pts · median 7.6e−13	**27.9 s** ×4.2 0.57 GiB peak · plan 2.70 MB **Rc 6.63 s** · ML 47.0 s DEV native 14.8 s ×2.2	**2.56 s** ×26 **OL 97.0 ms** · ML 247 ms · Rc 351 ms $B = 128$ fail ×20 inputs 99 % DEV native 45.8 ms ×0.47 pass ×0.36	**2.30 s** LC/FC 0.90 $B = 32$ · FC 2.55 s DEV native 50.7 ms · LC/FC 1.11
2 → 4	$u\bar{u} \to e^-e^+gg$ 6 legs · QCD · 69 families Rc·ML ≤ 1.1e−10 DEV 16 pts · median 6.4e−14 · †	*refused on loop* **Rc 833 ms** · ML 24.8 s DEV native 28.1 s ×34 · 0.37 GiB · 0.624 MB	*refused on loop: Z fermion signs (WP3.2), Furry zeros (WP3.5)* **Rc 10.2 ms** · ML 27.3 ms · OL no library DEV native 5.64 ms ×0.55 pass ×0.42 DEV numerator 49 %	*refused on loop*
	$gg \to t\bar{t}gg$ 6 legs · QCD · 1 086 families (LC 375) OL·Rc·ML ≤ 7.6e−9 DEV 16 pts · median 2.0e−11 · †	**62.5 s** ×1.0 0.86 GiB peak · plan 7.49 MB **Rc 61.4 s** · ML 218 s DEV native 36.6 s ×0.60	*no batched run on loop* G1 timed $B = 1$ only (Python inputs) · **OL 472 ms** · Rc 1.97 s · ML 5.45 s DEV native 190 ms ×0.40 pass ×0.31 DEV inputs 53 %	*not timed on loop* DEV native 231 ms · LC/FC 1.21
	$u\bar{u} \to gggg$ 6 legs · QCD · 628 families (LC 375) Rc 9.4e−12 · OL 1.9e−10 DEV 16 pts · median 1.5e−12 · †	**56.1 s** ×1.7 0.73 GiB peak · plan 8.48 MB **Rc 32.8 s** DEV native 39.4 s ×1.2	*no batched run on loop* G1 timed $B = 1$ only (Python inputs) · **OL 374 ms** · Rc 908 ms · ML not exported DEV native 190 ms ×0.51 pass ×0.39 DEV inputs 63 %	*not timed on loop* DEV native 247 ms · LC/FC 1.30
	$gg \to gggg$ 6 legs · QCD · 1 100 families (LC 550) ML·Rc·OL ≤ 4.0e−7 DEV 8 pts · median 3.1e−11 · †	**335 s** ×0.16 4.10 GiB peak · plan 45.4 MB **Rc 35.4 min** · ML 144.6 min DEV native 96.1 s ×0.045	*no batched run on loop* G1 timed $B = 1$ only (Python inputs) · **OL 2.84 s** · Rc 17.6 s · ML 38.7 s DEV native 985 ms ×0.35 pass ×0.27 DEV inputs 71 %	*not timed on loop* DEV native 1.09 s · LC/FC 1.10 · M: 1.02 s

	Process · validation	Generation	Full colour · per point, batched	Leading colour (LC)
	legs · corrections · loop families; largest deviation of the finite part per reference	generate + plan export, peak memory, plan size; provider generation on this host	median per point at the largest batch; provider times; gate G1 (time ÷ limit); stage shares	batched, same configuration; LC/FC; no provider ratio
2 → 5	$u\bar{u} \to e^-e^+d\bar{d}g$ 7 legs · QCD · families not recorded ML·Rc·OL ≤ 8.7e−13 DEV 3 pts	***refused on loop*** as the V+jets rows of G1; not run at this process **Rc 8.40 s** · ML 55.7 s DEV native 36.1 s ×4.3 · 0.44 GiB · 1.13 MB	***refused on loop: Z fermion signs (WP3.2), Furry zeros (WP3.5)*** as the V+jets rows of G1; not run at this process production, s/pt (20 Oct): OL 0.053–0.055 · Rc 0.035 · ML 0.143 gate limit 0.0455 s/pt (Recola) · seven-leg G1 rows not yet timed	***refused on loop***
	$u\bar{u} \to e^-e^+ggg$ 7 legs · QCD · families not recorded OL·ML·Rc ≤ 8.5e−13 DEV 3 pts	***refused on loop*** as the V+jets rows of G1; not run at this process **Rc 15.9 s** · ML 142 s DEV native 36.7 s ×2.3 · 0.48 GiB · 2.45 MB	***refused on loop: Z fermion signs (WP3.2), Furry zeros (WP3.5)*** as the V+jets rows of G1; not run at this process production, s/pt (20 Oct): OL 0.348–0.353 · Rc 0.246–0.259 · ML 1.41 gate limit 0.3198 s/pt (Recola) · seven-leg G1 rows not yet timed	***refused on loop***
	$gg \to t\bar{t}e^-e^+g$ 7 legs · QCD · families not recorded Rc·ML·OL ≤ 5.7e−9 DEV 3 pts	***not measured on loop*** **Rc 28.7 s** · ML 161 s DEV native 43.7 s ×1.5 · 0.59 GiB · 2.13 MB	***not measured*** production, s/pt (20 Oct): OL 0.425–0.433 · Rc 0.59–0.67 · ML 3.03–3.04 gate limit 0.5525 s/pt (OpenLoops) · seven-leg G1 rows not yet timed	***not timed on loop***
	$gg \to t\bar{t}e^-e^+g$ 7 legs · EW · families not recorded **not started on loop: EW loops (WP3.2)**	***not started on loop: EW loops (WP3.2)*** **Rc 21.9 s** · ML 10.9 min	***not started on loop: EW loops (WP3.2)*** production, s/pt (20 Oct): OL 0.96–1.12 · Rc 0.95–1.02 · ML 6.90–7.08 gate limit 1.235 s/pt (Recola) · seven-leg G1 rows not yet timed	***not started on loop***
	$u\bar{u} \to gggg$ 7 legs · QCD · families not recorded ML 2.2e−12 · Rc 2.9e−10 DEV 3 pts	***not measured on loop*** **Rc 170.3 min** · ML 12.2 h DEV native 517 s ×0.051 · 6.10 GiB · 69.2 MB	***not measured*** production, s/pt (20 Oct): OL no library · Rc not measured (protocol run, poles on, cache off: 180–184) · ML 569–570 no complete gate limit (Recola production not measured) · seven-leg G1 rows not yet timed	***not timed on loop***
	$gg \to gggg$ 7 legs · QCD · 9 414 families NJet DD 3.8e−9 DEV 3 pts · † · top-free (particle_veto = [t]), Nf = 5	***not measured on loop*** ML diagrams 17.4 h, stopped in export · Rc unfinished DEV native 54.9 min · 31.90 GiB · 160 MB	***not measured*** NJet 3.1.2 71.2 s per evaluation (one core; not matched) no matched provider: reported without a gate (addendum H)	***not timed on loop***

×1/16 ×1/4 ×1 ×4 ×16
faster pyAmpliCol ÷ fastest provider slower

RdBu, $\log_2$ scale, clamped. Grey: missing (reason given). White without a multiplier: no provider ratio.

Stage shares (bar; the largest is named): inputs numerator scatter NinjaRS masters glue (Table 4): inputs = trees, Born and counterterm inputs; glue = point parameters, input conversion, assembly. Times use the protocol's schedule B; "M:" gives the schedule G1 adopted for the class (Table 9). **References:** ML MadLoop 3.7.0, Rc Recola 2.2.5 (COLLIER), OL OpenLoops 2.1.5 (prebuilt libraries: no generation time), NJet DD NJet 3.1.2 in double-double. Deviation $|a-b|\,/\,\max(|a|,|b|,|B|\;\alpha_s/2\pi)$ of the binary64 finite part at the timed points (median: over the points, the largest over the references); † the largest lies within the binary64 stability estimate of the same configuration at that point. **Gate chip:** time ÷ binding limit (MadLoop, or 1.30 × the faster of Recola and OpenLoops); pass or fail when the whole uncertainty band is below or above 1, else open; ‡ $u\bar{u} \to u\bar{u}g$: pass (user decision 2026-10-24), measured open at 1.00, band 0.90–1.10.

Configurations. Main values: *integrated*, the engine on `loop` as timed by gate G1 at `1c971289` (LC: `d591a2f7`; both ancestors of `loop`; the later ct-simplify merge was not re-timed): Python per-point trees, Born and counterterm inputs, native loop stage (`loop-native-measure-g1-033edf15`). DEV in the G1 rows ($2 \to 2$ to $2 \to 4$): `work/v2-generation-scale` `889af02c` with native per-point inputs (timers `39824f9b`, `loop-native-measure-g1-genscale-39824f9b`); DEV in the $2 \to 5$ rows: the validations of `work/v2-generation-scale` with its production counterterm route, DESIGN.md WP3.1 eighth increment (0ea83af2); DESIGN.md WP3.1 seventh increment (fef92189) for $u\bar{u} \to gggg$ (generation, then single points). Generation excludes model loading. Unbatched development timings, such as a seven-gluon point with its rotation check, are not in this table (Section 4).

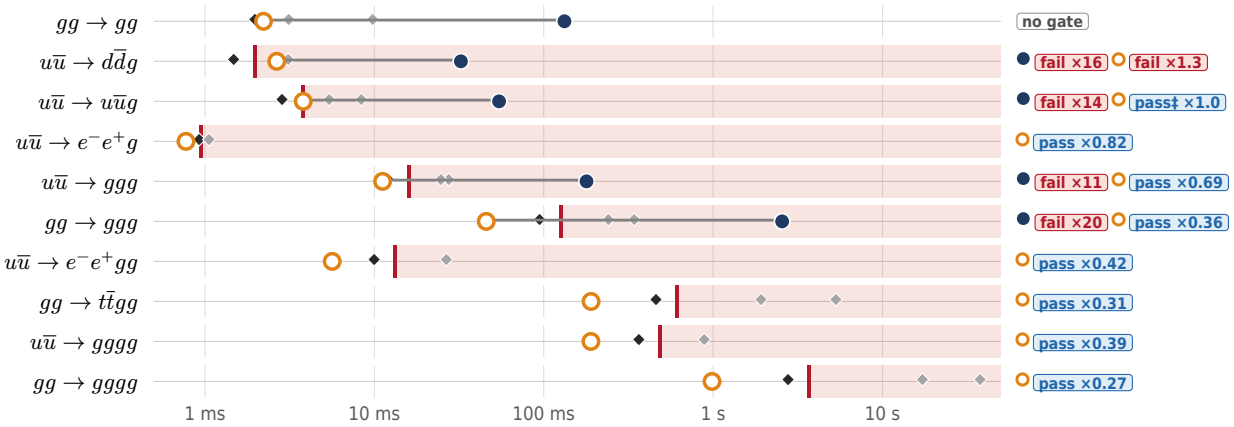


Figure 1: **Batched full colour against the gate** ($B = 128$ or the largest batch, finite part; the rows of Table 1 with a matched provider). Filled: the engine on `loop` (integrated; absent where G1 timed only $B = 1$ or the process is refused); open: native inputs (DEV); diamonds: the providers, the darkest the fastest; red line: the gate limit, the shaded side fails. Values: Table 1; the ratios and bands: Table 3.

1 Status at this milestone

On `loop` (engine b4d76cc3; merges since f132e53c)

`b71cf0e0`	Rust A1-A2	**Native loop-request DAG builder.**
`6c44c9f7`	WP2.7	**Gate G1 harness and measurement.**
`b4d76cc3`	addendum J	**Counterterms as plain tree insertions.**

Before (f132e53c): v2 waves 0-1, gates S1 and G0, native executor (WP2.1), colour (WP2.4), stability API (WP2.3), schedule M (WP2.8), tree-EW follow-up, NJet reference, observables (WP2.5), public v2 engine with plan v4 artifacts (WP2.6), counterterms in the recursion (WP2.2), runtime pole selection (WP2.10), Rust A0 (contract-class reuse pass), constant counterterm arguments (ct-assume-fix), addendum F-J, the previous report. **Native build** `loop-native-v2-slice-6c44c9f7` (digest `a9533f0e`…).

Gate G1 (Section 2; finite part, $2 \to 3$ and beyond)

- **On `loop` (integrated): not met.** It fails every gated process it generates: at $B = 128$ at ×11–20 of the binding limit (Python per-point inputs take 87 %–99 % of the time), and at ×12–27 for $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$, timed at $B = 1$ only. It refuses $u\bar{u} \to e^-e^+g$ and $u\bar{u} \to e^-e^+gg$ (Z fermion signs, WP3.2; Furry zeros, WP3.5).
- **Native inputs** DEV (WP3.1, not yet integrated) at $B = 128$: 8 of 9 pass ($u\bar{u} \to u\bar{u}g$ by user decision 2026-10-24 (1.00 of its limit, band 0.90–1.10)), $u\bar{u} \to d\bar{d}g$ fails (×1.35 of its limit). At $B = 1$ no process passes with 16 lanes (6 with one lane).
- **$2 \to 5$:** not timed yet (`measure-g1-7leg`); $u\bar{u} \to ggggg$ has no complete gate limit (Recola production not measured); $gg \to ggggg$ has no matched provider and is reported without a gate.

In development (branches, not on `loop`; HANDOFF §6)

- **Generation at scale** (WP3.1 G-int, `work/v2-generation-scale`): native trees, Born and counterterm inputs per point, native families and schedules, regeneration-free artifacts. Its validations are the DEV lines of the $2 \to 5$ rows of Table 1; integrating it is the next milestone, after which G1 is re-measured on the integrated configuration.
- **Rust ports** (HANDOFF §0.4): A0 and A1–A2 are on `loop` (not switched on); seeds (B) and cells (C) passed their phase-0 audits and wait for G-int; lowering (D) is in a fix round; A3 (counterterm-lowered model in the native DAG) is open.
- **Artifact portability** (partial branch) and the seven-leg G1 rows (`measure-g1-7leg`, specifications and streams).

Missing or incomplete

- Electroweak loops, Goldstones and the complex-mass scheme (WP3.2): $gg \to t\bar{t}e^-e^+g$ (EW) not started; V+jets refused on `loop`.
- Batched timings of the $2 \to 5$ set; LC timings of the V+jets rows; provider protocol-mode columns.
- Seven-gluon generation cost on the WP3.1 branch: 54.9 min with the plan export (generation 46.1 min, mostly the always-generated poles companion), peak 31.9 GiB (Section 4); inside addendum H, to be made lean.
- Ward check at seven gluons; prepared-artifact portability without regeneration (addendum G).

Performance gates and timing protocol: definitions

- **Gate.** Faster than MadLoop and at most 1.30× each matched Recola and OpenLoops time, on the **finite part**, from $2 \to 3$ upward; $2 \to 2$ is reported without a gate. Aspiration: fastest of all tools from $2 \to 3$.
- **Protocol.** Providers in production mode with poles off where they allow it (OpenLoops `truepoles 0`, Recola without IR poles, MadLoop production build); pyAmpliCol finite-only (`poles = "none"`) with stability checks off, one pinned thread, binary64, 16 lanes, the same frozen points. Statistic: median per point; the uncertainty band combines pyAmpliCol's block spread and the provider's repeat spread; a gate whose band contains the limit is **open**.
- **Timing boundary.** pyAmpliCol: the public v2 runtime call (`NativeExecutor.evaluate`) for a batch of B distinct points, everything included (per-point inputs, loop stage, assembly). Providers have no batch API: one time per point serves every B. Generation: pyAmpliCol `LoopSlice.generate` plus the plan export, one thread; MadLoop graph generation, export and compilation; Recola process generation; OpenLoops libraries are prebuilt (no generation time).
- **Labels.** Development measurements carry the branch and commit. In Table 1 they are secondary lines (DEV) and never a main value, a colour or a verdict chip of the main value; the detailed G1 tables colour both configurations, each labelled. Missing values are marked, never estimated or extrapolated; test durations are never used as event timings. The legacy per-diagram engine's timings of 18 October are history (data kept in `data/measurements.json`, not shown).

2 Gate G1: end-to-end timings of the v2 engine

G1 times the complete Born–virtual evaluation, **finite part only**, through the call of the public v2 runtime (`NativeExecutor.evaluate`, which `V2VirtualRuntime` calls) with stability checks off, on one pinned core and one thread, binary64, 16 SIMD lanes, in batches of B distinct points of a frozen stream (REQUIREMENTS_ADDENDUM D, plan §5.8). Two pyAmpliCol configurations are timed:

- **integrated** — this report's branch at `1c971289` (LC rows at `d591a2f7`, the same engine code): the engine on `loop` since 24 October plus the native-DAG merge (no production call site switched) and this package's timers. Trees, Born amplitudes and counterterm scalars are formed per point in Python, then the loop stage runs natively. It refuses the V+jets rows: their closed-quark Agg triangles need Furry's certified zeros (WP3.5), and their Z couplings on loop quark lines the electroweak fermion signs (WP3.2);

- **native inputs** development — work/v2-generation-scale 889af02c (WP3.1, under integration; scratch commit 39824f9b adds the timers), where the executor also forms the trees, the Born and the counterterm inputs. It generates the V+jets rows with γ and Z exchange.

The providers run in production mode on the same stream: OpenLoops with truepoles 0, Recola without IR poles (one amplitude call per point), MadLoop at its production stability setting on exports built with MG5′s production flags. OpenLoops has no installed library with QCD corrections to $\ell^+\ell^-$ + 1 or 2 jets (pplj_ew carries electroweak corrections only; existing provider builds only), so the V+jets gates bind on MadLoop and Recola. Providers have no batch API: one time per point serves every B. **Gate statistic:** the median per point on both sides — over pyAmpliCol's timed blocks of one B, and over the provider's timed points (the stream's first 16; 8 for MadLoop $gg \to gggg$, Recola $gg \to gggg$, OpenLoops $gg \to gggg$ and Recola $u\bar{u} \to gggg$) in its run on the core that timed the native-input configuration (other cores: MadLoop $gg \to gggg$, Recola $gg \to gggg$ and Recola $u\bar{u} \to gggg$). Its uncertainty is pyAmpliCol's block spread (half the range of three blocks, the median's standard error from five) plus the provider's half-range over its repeated runs (8 %, the campaign's largest, for the single runs of Recola $gg \to gggg$ and Recola $u\bar{u} \to gggg$). A gate passes or fails only if the whole band does; otherwise it is **open** and the configuration is repeated once (plan §5.8 item 10). Every timed call returned all its points (failed points: 0); rescue is off by protocol (no checks, no escalation), so the rescue rate is 0.

Table 2: **Gate G1, full colour**, ms per point (gate statistic: median), finite part, one thread. Rows $B = 1$ and $B = 128$ (the integrated $2 \to 4$ rows only at $B = 1$: their Python inputs take seconds per point). Multiplier and colour: time ÷ the fastest provider of the row (legend of Table 1). Grey rows: generation — pyAmpliCol generation plus plan export with its peak memory; MadLoop graph generation, export and compilation · the share of its loop-numerator coefficients in its time (Table 4); Recola process generation; OpenLoops: the prebuilt library and its load plus first call. MadLoop $u\bar{u} \to gggg$ was not exported: the $gg \to gggg$ export took 2.4 h here (HANDOFF §0.1), and OpenLoops binds the gate. Gate: upper line integrated, lower line native inputs; the number is the time over the binding limit (MadLoop's time or 1.30× the faster of Recola and OpenLoops); **open**: the band contains the limit; ‡ pass by user decision although it does (bands: Table 3).

Process	B	Integrated	Native inputs^{dev}	MadLoop	Recola	OpenLoops	Gate
$gg \to gg$ $2 \to 2$, no gate	1	**140** ×69	**14.7** ×7.3	**2.02** fastest	**10.0** ×5.0	**3.20** ×1.6	no gate
	128	**132** ×66	**2.22** ×1.1	**2.02** fastest	**10.0** ×5.0	**3.20** ×1.6	no gate
	gen.	10.4 s · 0.3 GiB	9.82 s · 0.3 GiB	9.70 s · 64 %	0.142 s	ppjj · 6.80 ms	
$u\bar{u} \to d\bar{d}g$ $2 \to 3$	1	**40.6** ×27	**10.9** ×7.2	**3.17** ×2.1	**3.18** ×2.1	**1.52** fastest	**fail** ×21 **fail** ×5.5
	128	**32.4** ×21	**2.65** ×1.8	**3.17** ×2.1	**3.18** ×2.1	**1.52** fastest	**fail** ×16 **fail** ×1.3
	gen.	9.94 s · 0.3 GiB	10.6 s · 0.3 GiB	9.87 s · 73 %	0.406 s	ppjjj · 4.73 ms	
$u\bar{u} \to u\bar{u}g$ $2 \to 3$	1	**68.9** ×24	**19.1** ×6.5	**8.59** ×2.9	**5.55** ×1.9	**2.92** fastest	**fail** ×18 **fail** ×5.0
	128	**54.5** ×19	**3.80** ×1.3	**8.59** ×2.9	**5.55** ×1.9	**2.92** fastest	**fail** ×14 **pass‡** 1.00
	gen.	10.4 s · 0.3 GiB	10.9 s · 0.3 GiB	10.9 s · 75 %	0.462 s	ppjjj · 8.43 ms	
$u\bar{u} \to e^-e^+g$ $2 \to 3$	1	*not generable*	**3.62** ×3.8	**0.948** fastest	**1.08** ×1.1	*no library*	— **fail** ×3.8
	128	*not generable*	**0.772** ×0.82	**0.948** fastest	**1.08** ×1.1	*no library*	— **pass** 0.82
	gen.	CT coverage refused	26.0 s · 0.4 GiB	14.0 s · 79 %	0.701 s	—	
$u\bar{u} \to ggg$ $2 \to 3$	1	**215** ×17	**50.6** ×4.1	**28.2** ×2.3	**25.3** ×2.0	**12.4** fastest	**fail** ×13 **fail** ×3.1
	128	**179** ×14	**11.2** ×0.90	**28.2** ×2.3	**25.3** ×2.0	**12.4** fastest	**fail** ×11 **pass** 0.69
	gen.	12.3 s · 0.4 GiB	12.2 s · 0.3 GiB	17.6 s · 76 %	0.634 s	ppjjj · 23.7 ms	
$gg \to ggg$ $2 \to 3$	1	**2 760** ×28	**211** ×2.2	**247** ×2.5	**351** ×3.6	**97.0** fastest	**fail** ×22 **fail** ×1.7
	128	**2 560** ×26	**45.8** ×0.47	**247** ×2.5	**351** ×3.6	**97.0** fastest	**fail** ×20 **pass** 0.36
	gen.	27.9 s · 0.6 GiB	14.8 s · 0.4 GiB	47.0 s · 78 %	6.63 s	ppjjj · 159 ms	
$u\bar{u} \to e^-e^+gg$ $2 \to 4$	1	*not generable*	**43.3** ×4.2	**27.3** ×2.7	**10.2** fastest	*no library*	— **fail** ×3.3
	128	*not generable*	**5.64** ×0.55	**27.3** ×2.7	**10.2** fastest	*no library*	— **pass** 0.42
	gen.	CT coverage refused	28.1 s · 0.4 GiB	24.8 s · 83 %	0.833 s	—	
$gg \to gggg$ $2 \to 4$	1	**101 000** ×36	**4 100** ×1.4	**38 700** ×14	**17 600** ×6.2	**2 840** fastest	**fail** ×27 **fail** ×1.1
	128	*not timed*	**985** ×0.35	**38 700** ×14	**17 600** ×6.2	**2 840** fastest	— **pass** 0.27
	gen.	335 s · 4.1 GiB	96.1 s · 2.0 GiB	8 680 s · 51 %	2 130 s	ppjjjj · 4 950 ms	
$gg \to t\bar{t}gg$ $2 \to 4$	1	**7 310** ×15	**840** ×1.8	**5 450** ×12	**1 970** ×4.2	**472** fastest	**fail** ×12 **fail** ×1.4
	128	*not timed*	**190** ×0.40	**5 450** ×12	**1 970** ×4.2	**472** fastest	— **pass** 0.31
	gen.	62.5 s · 0.9 GiB	36.6 s · 0.7 GiB	218 s · 64 %	61.4 s	pptjj · 821 ms	

Process	B	Integrated	Native inputs^{dev}	MadLoop	Recola	OpenLoops	Gate
$u\bar{u} \to gggg$ $2 \to 4$	1	**5 760** ×15	**860** ×2.3	*not timed*	**908** ×2.4	**374** fastest	**fail** ×12 **fail** ×1.8
	128	*not timed*	**190** ×0.51	*not timed*	**908** ×2.4	**374** fastest	— **pass** 0.39
	gen.	56.1 s · 0.7 GiB	39.4 s · 0.7 GiB	—	32.8 s	ppjjjj · 606 ms	

Table 3: **The gate at every batch size** (native inputs^{dev}, full colour): time over the binding limit and its uncertainty band (cell colour: blue pass, red fail, amber open; ‡ pass by user decision, the band containing the limit: $u\bar{u} \to u\bar{u}g$ by user decision 2026-10-24 (1.00 of its limit, band 0.90–1.10)). $B = 1$ is timed with the default 16 lanes (a one-point call pays a 16-lane chunk) and with one lane. The $u\bar{u} \to u\bar{u}g$ and $u\bar{u} \to d\bar{d}g$ rows are the single repeat; their first runs gave $u\bar{u} \to u\bar{u}g$ 3.80 ms and $u\bar{u} \to d\bar{d}g$ 2.63 ms at $B = 128$.

Process	$B = 1$, 16 lanes		$B = 1$, 1 lane		$B = 8$		$B = 32$		$B = 128$	
$u\bar{u} \to d\bar{d}g$	**5.54**	5.46–5.62	**2.13**	2.10–2.16	**1.46**	1.37–1.55	**1.15**	1.12–1.17	**1.35**	1.19–1.51
$u\bar{u} \to u\bar{u}g$	**5.03**	4.93–5.14	**1.63**	1.59–1.67	**1.15**	1.07–1.22	**0.87**	0.85–0.90	**1.00‡**	0.90–1.10
$u\bar{u} \to e^-e^+g$	**3.82**	3.81–3.83	**1.61**	1.60–1.62	**1.07**	1.01–1.13	**0.85**	0.83–0.87	**0.82**	0.81–0.82
$u\bar{u} \to ggg$	**3.14**	3.06–3.22	**0.84**	0.81–0.86	**0.68**	0.66–0.71	**0.74**	0.61–0.87	**0.69**	0.64–0.75
$gg \to ggg$	**1.67**	1.67–1.67	**0.45**	0.45–0.45	**0.42**	0.40–0.44	**0.34**	0.33–0.35	**0.36**	0.36–0.36
$u\bar{u} \to e^-e^+gg$	**3.26**	3.24–3.28	**0.78**	0.78–0.78	**0.62**	0.21–1.02	**0.42**	0.41–0.42	**0.42**	0.40–0.45
$gg \to gggg$	**1.11**	1.10–1.12	**0.31**	0.30–0.31	**0.30**	0.30–0.31	**0.26**	0.24–0.29	**0.27**	0.25–0.28
$gg \to t\bar{t}gg$	**1.37**	1.35–1.38	**0.37**	0.36–0.37	**0.35**	0.33–0.37	**0.28**	0.27–0.29	**0.31**	0.30–0.32
$u\bar{u} \to gggg$	**1.77**	1.74–1.80	**0.43**	0.43–0.44	**0.47**	0.43–0.51	**0.37**	0.32–0.41	**0.39**	0.37–0.41

2.1 Where the time goes

Table 4: **Stage split**, ms per point, full colour, at the largest batch timed (integrated $2 \to 4$: $B = 1$). Trees/Born/CT: the per-point inputs — in Python for the integrated slice (trees, counterterm program, Born and counterterm colour projections, scalars), in the executor with native inputs. Numerator: seeds, fused factors and the dressing of the open-loop objects (the pure numerator evaluation). Scatter: closures into NinjaRS lanes. NinjaRS: the reduction without the masters; masters: the master integrals (canonical arguments, cache, oneloopmaster). Glue: point parameters, carrier conversion, assembly and the rest of the Python call. MadLoop (instrumented copy; its points less one, 3 for $gg \to gggg$): Born/counterterm HELAS calls, loop-numerator coefficients, reduction plus masters, residual (stability test, colour sums). Bar: shares, colours as the column heads.

Process		Trees, Born, CT	Numerator	Scatter	NinjaRS	Masters	Glue	Total	Shares
$gg \to gg$	native^{dev}	0.941	0.836	0.0886	0.114	0.0378	0.206	**2.22**	
	integrated	128	0.851	0.0919	0.119	0.0381	2.96	**132**	
	MadLoop	0.0315	1.27	—	0.585		0.107	**1.99**	
$u\bar{u} \to d\bar{d}g$	native^{dev}	0.405	0.753	0.464	0.139	0.112	0.785	**2.65**	
	integrated	28.1	0.749	0.462	0.137	0.111	2.66	**32.4**	
	MadLoop	0.0479	2.35	—	0.683		0.133	**3.22**	
$u\bar{u} \to u\bar{u}g$	native^{dev}	0.768	1.22	0.472	0.223	0.161	0.970	**3.80**	
	integrated	48.5	1.23	0.479	0.232	0.171	3.99	**54.5**	
	MadLoop	0.137	6.44	—	1.15		0.814	**8.54**	
$u\bar{u} \to e^-e^+g$	native^{dev}	0.128	0.266	0.151	0.0266	0.0179	0.182	**0.772**	
	MadLoop	0.0184	0.754	—	0.133		0.0460	**0.951**	
$u\bar{u} \to ggg$	native^{dev}	5.85	2.94	0.480	0.468	0.228	0.535	**11.2**	
	integrated	168	2.92	0.476	0.459	0.225	5.75	**179**	
	MadLoop	0.346	18.5	—	2.54		2.87	**24.3**	
$gg \to ggg$	native^{dev}	28.3	11.8	0.538	2.05	0.487	2.76	**45.8**	
	integrated	2 530	11.7	0.519	2.03	0.483	10.5	**2 560**	
	MadLoop	2.42	184	—	11.3		39.5	**237**	
$u\bar{u} \to e^-e^+gg$	native^{dev}	1.13	2.75	0.637	0.423	0.233	0.459	**5.64**	
	MadLoop	0.289	22.7	—	2.38		2.11	**27.5**	
$gg \to gggg$	native^{dev}	696	221	2.42	47.9	5.85	5.35	**985**	
	integrated	97 300	3 750	4.08	38.6	5.81	137	**101 000**	
	MadLoop	88.9	18 600	—	209		17 700	**36 500**	
$gg \to t\bar{t}gg$	native^{dev}	102	45.9	0.923	20.4	6.31	15.0	**190**	
	integrated	6 490	751	3.55	20.7	6.58	29.3	**7 310**	
	MadLoop	24.2	3 290	—	105		1 740	**5 160**	
$u\bar{u} \to gggg$	native^{dev}	119	47.9	1.81	9.14	2.90	8.49	**190**	
	integrated	4 910	806	2.01	9.64	2.99	24.1	**5 760**	

Verdict of gate G1 (processes from $2 \to 3$; $2 \to 2$ without a gate).

- **Integrated slice: fails** every process it generates: at $B = 128$ at ×11–20 of the binding limit, the per-point Python inputs taking 87 %–99 % of its time (Table 4); at $B = 1$ at ×12–27, where $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$ were timed only. It does not generate $u\bar{u} \to e^-e^+g$ and $u\bar{u} \to e^-e^+gg$ (counterterm coverage: Furry zeros, WP3.5; Z fermion signs, WP3.2).
- **Native inputs** development at $B = 128$: **pass** for $u\bar{u} \to u\bar{u}g$, $u\bar{u} \to e^-e^+g$, $u\bar{u} \to ggg$, $gg \to ggg$, $u\bar{u} \to e^-e^+gg$, $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$ ($u\bar{u} \to u\bar{u}g$ by user decision 2026-10-24 (1.00 of its limit, band 0.90–1.10)); **fail** for $u\bar{u} \to d\bar{d}g$ (×1.3, OpenLoops binding). The gate holds for every B (Table 3): at $B = 1$ with 16 lanes none passes (a one-point call pays a 16-lane chunk); with one lane per call $u\bar{u} \to ggg$, $gg \to ggg$, $u\bar{u} \to e^-e^+gg$, $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$ pass.
- **Fastest of all tools** (aspiration), native inputs at $B = 128$, within the uncertainty band: faster than every provider for $u\bar{u} \to e^-e^+g$, $u\bar{u} \to ggg$, $gg \to ggg$, $u\bar{u} \to e^-e^+gg$, $gg \to gggg$, $gg \to t\bar{t}gg$ and $u\bar{u} \to gggg$; at parity with the fastest for none; slower than the fastest (OpenLoops) for $u\bar{u} \to d\bar{d}g$ and $u\bar{u} \to u\bar{u}g$. NJet, not a matched provider, takes 4.58 s for its two-evaluation full-colour $gg \to gggg$ test (Table 13).
- Seven-leg rows ($u\bar{u} \to e^-e^+d\bar{d}g$, $u\bar{u} \to e^-e^+ggg$, $gg \to t\bar{t}e^-e^+g$, $u\bar{u} \to ggggg$, $gg \to ggggg$) need the WP3.1 generator and are added by the follow-on package `measure-g1-7leg`; $gg \to ggggg$ has no matched provider timing (addendum F, H): reported without a gate.

Table 5: **Levers**, measured on the rows of Table 4. Gains are measured where a column says so; otherwise they are the stage's share, an upper bound.

Lever	Evidence	Expected gain	Owner
Native per-point inputs (WP3.1 I3–I6: trees, Born, counterterm scalars in the executor)	Python inputs 87 %–99 % of the integrated time at $B = 128$	measured, integrated ÷ native at equal B: ×3.6–25 at $B = 1$, ×12–56 at $B = 128$ ($2 \to 3$)	`generation-scale` (G-int, branch; audit pending)
Compiled native trees (kernels as the tree engine's SymJIT planes, lanes batched)	native trees 15 %–71 % of the native-input time; $gg \to gggg$ 71 %	up to the share; ×3.4 for the largest	Rust ports A/B (`native-dag`, HANDOFF §0.4)
Lanes by batch in binary64 (`plan_export.lanes_for`: 16 lanes also for one point)	$B = 1$: one 16-lane chunk per call	measured: ×2.4–4.2 at $B = 1$ (16 lanes ÷ 1 lane)	executor policy (WP3.1 or wave 4)
Finite-only result path (sum the virtual natively, no per-family lane export, Python assembly only for checks and observables)	glue 1 %–30 % of the native-input time	up to the share (largest at $2 \to 3$ with quarks)	`plan_export`, executor (wave 4)
Lane-major closures (no scatter, no second copy into the reducer's batch)	scatter 0 %–20 %	up to the share	executor, reducer adapter (wave 4)
AVX2 / AVX-512 kernels (runtime selection)	numerator 22 %–49 %	×1.0–1.4 on the dressing (gate G0)	wave 4 (WP4.6)
NinjaRS and masters: parallel reduction, transposed reduction, stage-2 sharing, a cheaper master cache key	NinjaRS + masters 5 %–14 %; masters 11 %–45 % of the reduction, oneloopmaster alone 2 %–12 %: the cache bookkeeping costs more than the integrals	one thread: up to the share; threads: DESIGN §5.6	NinjaRS requests (DESIGN §11.5)

The failing native-input row, $u\bar{u} \to d\bar{d}g$, takes 2.65 ms at $B = 128$ against a limit of 1.97 ms (OpenLoops 1.52 ms × 1.30). Its glue and scatter are 1.25 ms; without them it would take 1.40 ms, 0.71 of the limit. At $B = 32$ it takes 2.26 ms: its glue grows with the batch (0.325 ms per point at $B = 32$, 0.785 ms at $B = 128$).

2.2 Leading colour

Table 6: **Leading colour** (LC), ms per point, same protocol as Table 2; in each cell the full-colour (FC) time of the same configuration and batch, and LC/FC. No provider ratio: the providers time full colour. LC uses the committed QCD colour continuation (`examples/loop/source_colour`); the V+jets rows were not timed at LC.

Process	Integrated				Native inputs[dev]			
	$B = 1$		largest batch		$B = 1$		largest batch	
$u\bar{u} \to d\bar{d}g$	**40.5**	FC 40.6 · 1.00	**37.4**	FC 32.4 · 1.15	**5.25**	FC 10.9 · 0.48	**1.91**	FC 2.65 · 0.72
$u\bar{u} \to u\bar{u}g$	*not timed*		*not timed*		**9.00**	FC 19.1 · 0.47	**3.16**	FC 3.80 · 0.83
$u\bar{u} \to ggg$	**184**	FC 215 · 0.86	**171**	FC 179 · 0.96	**22.8**	FC 50.6 · 0.45	**12.6**	FC 11.2 · 1.13
$gg \to ggg$	**2 350**	FC 2 760 · 0.85	**2 300** ($B = 32$)	FC 2 550 · 0.90	**91.1**	FC 211 · 0.43	**50.7**	FC 45.8 · 1.11
$gg \to gggg$	*not timed*		*not timed*		**2 070**	FC 4 100 · 0.50	**1 090**	FC 985 · 1.10
$gg \to t\bar{t}gg$	*not timed*		*not timed*		**418**	FC 840 · 0.50	**231**	FC 190 · 1.21
$u\bar{u} \to gggg$	*not timed*		*not timed*		**380**	FC 860 · 0.44	**247**	FC 190 · 1.30

LC is cheaper in the loop stage (fewer families and skeletons), but not yet end to end: with native inputs the trees at LC cost more than at full colour for the processes with many colour orderings ($u\bar{u} \to ggg$: 9.93 against 5.85 ms per point), and in the integrated slice the Python inputs dominate either way.

2.3 Throughput, model recalibration and schedule M

Table 7: **Kernel throughput and the design recipe** (native inputs, full colour, largest batch). Work: the executor's modelled complex multiply-adds of the chosen kernel forms per point (one lane); throughput: work × lanes over the dressing time (fused-factor builds excluded). Native loop: seeds, factors, dressing, scatter and reduction; outside: everything else of the call (trees, glue). Recipe: DESIGN §1.3, finite only, where the design has a row; measured ÷ recipe for the total and for the NinjaRS stage (masters included) against its model R.

Process	Work, MCMAC/pt	Dressing, ms	G CMAC/s	Native loop, ms	Outside, ms	Total, ms	Total ÷ recipe	NinjaRS ÷ R
$gg \to gg$	0.932	0.694	1.34	1.08	1.14	**2.22**	1.5–4.4	0.97–1.10
$u\bar{u} \to d\bar{d}g$	0.556	0.446	1.25	1.47	1.19	**2.65**	no row	—
$u\bar{u} \to u\bar{u}g$	1.09	0.864	1.26	2.08	1.73	**3.80**	no row	—
$u\bar{u} \to e^-e^+g$	0.168	0.131	1.28	0.462	0.311	**0.772**	no row	—
$u\bar{u} \to ggg$	3.18	2.44	1.30	4.12	7.07	**11.2**	no row	—
$gg \to ggg$	14.3	10.9	1.32	14.9	30.9	**45.8**	no row	—
$u\bar{u} \to e^-e^+gg$	2.89	2.22	1.30	4.05	1.59	**5.64**	no row	—
$gg \to gggg$	268	215	1.25	278	708	**985**	2.4–7.1	1.05–1.24
$gg \to t\bar{t}gg$	56.0	43.2	1.29	73.5	117	**190**	1.2–3.5	0.99–1.39
$u\bar{u} \to gggg$	56.7	43.6	1.30	61.8	128	**190**	no row	—

Recalibration. The kernels run 1.25–1.34 G complex MAC/s per core on the real plans, the rate gate G0 measured on its synthetic programs (1.29–1.37); the dressing term of the recipe holds. The NinjaRS stage, masters included, lands on the design's model R (measured ÷ R 0.97–1.39, Table 7). What the recipe misses is the **outer factor**: it assumed trees, seeds and counterterms add 5–10 % (DESIGN §1.3), but with the current native inputs the total is 1.66–3.6 times dressing plus reduction at full colour (up to 13 at LC). Table 8 re-estimates every design row with these measured terms.

Table 8: **Design estimates recalibrated by G1** (`tools/loop_v2_costmodel.py --g1`), ms per point, finite only, one thread. Recipe: DESIGN Table 1a's finite-only band (outer factor 1.05–1.10). G1: the recipe's dressing term, the NinjaRS model R × 0.97–1.39 (measured above) and the measured outer factor of the row's class representative at $B = 128$ with native inputs (same colour where measured; LC rows of the V+jets class take the FC factor, marked †). Measured: G1's own time where the row was timed. Gate: 1.3× the best production provider of DESIGN Table 1b (seven-leg rows, protocol of 20 October), with the headroom gate ÷ upper estimate. The outer factor shrinks with compiled trees (Table 5); until then these replace the Table 1a targets.

Design row	Col.	Recipe	Outer	Class	G1 estimate	Measured	Gate	Headroom
$gg \to gg$	FC	0.502–1.52	2.25	$gg \to gg$	**1.07–3.25**	2.22	—	—
$gg \to gg$	LC	0.204–0.542	2.25†	$gg \to gg$	**0.433–1.18**	—	—	—
$gg \to gggg$	FC	139–409	3.61	$gg \to gggg$	**474–1 410**	985	3 800	2.69
$gg \to gggg$	LC	55.7–146	11.03	$gg \to gggg$	**578–1 570**	1 090	—	—
$gg \to ggggg$	FC	2 710–7 920	3.61	$gg \to gggg$	**9 210–27 400**	—	—	—
$gg \to ggggg$	LC	1 070–2 830	11.03	$gg \to gggg$	**11 100–30 500**	—	—	—
$u\bar{u} \to ggggg$	FC	827–2 630	3.30	$u\bar{u} \to gggg$	**2 580–8 260**	—	—	—
$u\bar{u} \to ggggg$	LC	206–526	12.66	$u\bar{u} \to gggg$	**2 440–6 670**	—	—	—
$u\bar{u} \to e^-e^+ggg$	FC	24.5–68.1	1.66	$u\bar{u} \to e^-e^+gg$	**38.3–111**	—	320	2.88
$u\bar{u} \to e^-e^+ggg$	LC	3.34–8.31	1.66†	$u\bar{u} \to e^-e^+gg$	**5.21–13.9**	—	—	—
$u\bar{u} \to e^-e^+d\bar{d}g$	FC	5.34–16.5	1.66	$u\bar{u} \to e^-e^+gg$	**8.34–27.1**	—	45.1	1.67
$u\bar{u} \to e^-e^+d\bar{d}g$	LC	1.19–3.14	1.66†	$u\bar{u} \to e^-e^+gg$	**1.85–5.32**	—	—	—
$gg \to t\bar{t}gg$	FC	54.8–156	2.68	$gg \to t\bar{t}gg$	**138–408**	190	633	1.55
$gg \to t\bar{t}gg$	LC	14.6–40.1	11.06	$gg \to t\bar{t}gg$	**152–434**	231	—	—
$gg \to t\bar{t}e^-e^+g$	FC	27.4–76.1	2.68	$gg \to t\bar{t}gg$	**69.0–200**	—	553	2.76
$gg \to t\bar{t}e^-e^+g$	LC	4.06–11.0	11.06	$gg \to t\bar{t}gg$	**42.2–119**	—	—	—

Table 9: **Schedule M against B** (native inputs, one thread), pure-gluon classes, the only classes where M merges seed entries (DESIGN §5.6). The schedule-dependent stages are seeds, factors, dressing, scatter and the NinjaRS reduction ("the reduction included", §1.6 item 5); the trees, Born and counterterm inputs and the glue are the same work in both schedules. M is adopted where M/B ≤ 0.85 on the schedule-dependent stages at both batch sizes; ± is the timing uncertainty of the ratio.

Process	Col.	Stages M/B, $B = 1$	Stages M/B, $B = 128$	End to end, $B = 1$	End to end, $B = 128$	Adopt M
$gg \to ggg$	FC	**0.99** ± 0.00	**1.00** ± 0.03	0.99	1.11	no
$gg \to ggg$	LC	**0.90** ± 0.00	**0.93** ± 0.05	0.93	0.92	no
$gg \to gggg$	FC	**0.88** ± 0.01	**0.85** ± 0.09	0.89	0.92	no
$gg \to gggg$	LC	**0.75** ± 0.02	**0.79** ± 0.01	0.86	0.94	**yes**

B against M (decision, provisional until the trees are compiled). On the schedule-dependent stages M passes the 15 % rule for $gg \to gggg$ at LC (as DESIGN §5.6 measured on the dressing side, 0.75–0.80), and not for the others: $gg \to gggg$ at FC lies at 0.85–0.88, the rule's edge within its uncertainty, and $gg \to ggg$ at 0.90–1.0. End to end the gain is

diluted by the tree inputs, which both schedules share and which compiled trees will shrink (Table 5). With a quark line M's entries do not merge at all (M/B work 1.00, DESIGN §5.6; equal by test). **Decision (G1):** schedule M for pure-gluon classes at leading colour from $n = 6$; schedule B for every other class at $n \leq 6$, including pure-gluon full colour; M stays the planned default from $n \geq 8$, and both are re-measured at seven legs with the WP3.1 plans (`measure-g1-7leg`).

Table 10: **Values at the timed points**: pyAmpliCol's binary64 finite part (native inputs) against each provider's production-mode value on the common stream points (the providers read the binary64 decimal text of the exact points), largest deviation, metric $|a - b| \,/\, \max(|a|, |b|, |B|\ \alpha_s/2\pi)$; Recola's finite part converted with the universal double pole (it returns no poles in production). Worst point: the point of the row's largest deviation, evaluated again with one rotation and the reduction-direction check in binary64 and in double-double (DD): E is the binary64 estimate, DD the DD value's deviation from the closest provider (named). Campaign: the first three points against the committed references (`tests/fixtures/loop/required`), fixture metric and tolerance, native inputs and integrated.

Process	Pts	vs MadLoop	vs Recola	vs OpenLoops	Worst point: E	DD: closest provider	Campaign, native	Campaign, integr.
$gg \to gg$	16	1.7×10^{-12}	7.4×10^{-13}	7.5×10^{-13}	—	—	—	—
$u\bar{u} \to d\bar{d}g$	16	6.8×10^{-13}	3.3×10^{-10}	6.2×10^{-13}	pt 1: 7.2×10^{-12}	3.2×10^{-13} MadLoop	3.6×10^{-10}	3.6×10^{-10}
$u\bar{u} \to u\bar{u}g$	16	3.4×10^{-12}	7.2×10^{-11}	6.8×10^{-12}	pt 1: 9.2×10^{-12}	2.4×10^{-13} MadLoop	8.6×10^{-11}	8.6×10^{-11}
$u\bar{u} \to e^-e^+g$	16	8.8×10^{-14}	4.9×10^{-14}	no library	pt 0: 1.4×10^{-12}	7.5×10^{-15} Recola	—	—
$u\bar{u} \to ggg$	16	1.4×10^{-12}	1.5×10^{-12}	8.4×10^{-11}	pt 2: 1.2×10^{-12}	1.3×10^{-14} MadLoop	—	—
$gg \to ggg$	16	1.7×10^{-6}	1.7×10^{-6}	1.7×10^{-6}	pt 10: 9.9×10^{-6}	6.0×10^{-12} Recola	—	—
$u\bar{u} \to e^-e^+gg$	16	1.1×10^{-10}	5.0×10^{-11}	no library	pt 8: 2.5×10^{-9}	4.3×10^{-13} Recola	—	—
$gg \to gggg$	8	1.7×10^{-7}	4.0×10^{-7}	4.0×10^{-7}	pt 1: 4.8×10^{-6}	4.4×10^{-12} Recola	4.0×10^{-7} **above 2.0×10^{-8}**	8.7×10^{-8} **above 2.0×10^{-8}**
$gg \to t\bar{t}gg$	16	7.6×10^{-9}	1.5×10^{-9}	1.5×10^{-9}	pt 13: 5.6×10^{-8}	5.5×10^{-15} OpenLoops	1.1×10^{-11}	1.1×10^{-11}
$u\bar{u} \to gggg$	16	—	9.4×10^{-12}	1.9×10^{-10}	pt 9: 2.7×10^{-9}	1.9×10^{-13} OpenLoops	—	—

Reading the values. The Born agrees with every provider to 1.0×10^{-13} or better. For the finite part, the double-double (DD) value at each worst point (Table 10) separates the sources: it agrees with the closest provider to 7.3×10^{-12} or better everywhere. Where pyAmpliCol's binary64 estimate E exceeds the deviation — $gg \to ggg$, $gg \to gggg$ and $gg \to t\bar{t}gg$ (above 10^{-9}) and $u\bar{u} \to gggg$ (1.9×10^{-13} in DD against 1.9×10^{-10} in binary64) — the deviation is pyAmpliCol's own binary64 loss, which E flags. The other deviations above 10^{-11} stay in DD and are the providers' own accuracy: Recola's at $u\bar{u} \to d\bar{d}g$ (3.3×10^{-10}), $u\bar{u} \to u\bar{u}g$ and $u\bar{u} \to e^-e^+gg$ (COLLIER's default target 10^{-8}, HANDOFF §7), OpenLoops' at $u\bar{u} \to ggg$ (8.4×10^{-11}) and MadLoop's at $u\bar{u} \to e^-e^+gg$ (1.1×10^{-10}); MadLoop's production values also stay off the DD values at the ill-conditioned gluon and top points (up to 2.4×10^{-7}). At the campaign points $gg \to gggg$ point 1 misses the fixture tolerance (2×10^{-8}) in binary64 on both paths, within its estimate $E = 4.8\times10^{-6}$; its DD value agrees with Recola to 4.4×10^{-12}.

3 Required processes: references and development validation

The plan requires the $2 \to 5$ processes of Table 1 (plan §5.7), compared with MadLoop and Recola, and with OpenLoops where a public library exists. A reference provider is run only if its generation completes in about 45 minutes, or in a time comparable to pyAmpliCol's own generation of the process (HANDOFF §0.1, addendum F). Otherwise pyAmpliCol is validated by Ward and gauge identities, Catani poles, LC/NLC/FC and DD/AP consistency, stability checks, lower multiplicities and existing references.

Table 11: **Provider generation on this host and the 45-minute rule** (HANDOFF §0.1): the rows of Table 1 where a provider generation takes more than 45 minutes or did not finish. MadLoop: graph generation + source export + compilation; Recola: process generation; OpenLoops: prebuilt public libraries (no generation time). Records: `data/g1.json` (G1 rows) and `data/measured.json` (`provider_generation`, the reference runs of the $2 \to 5$ rows), through `data/summary.json`. Every other row of Table 1 generates within 45 minutes with every provider (longest: MadLoop 10.9 min for $gg \to t\bar{t}e^-e^+g$ (EW)).

Process	MadLoop	Recola	OpenLoops	45-minute rule
$gg \to gggg$	2.4 h (285 + 8 354 + 37 s)	35.4 min	prebuilt ppjjjj	**MadLoop over 45 min; results exist and are used**
$u\bar{u} \to ggggg$	12.2 h (5 793 + 38 040 + 122 s)	2.8 h	no public library	**MadLoop, Recola over 45 min; results exist and are used**
$gg \to ggggg$	diagrams 17.4 h; unfinished (last state: exporting)	unfinished (last state: running, 2026-10-20)	no public library	**stopped 22 Oct; not to be restarted (HANDOFF section 0.1)**

Table 12: **Native validation of the required set** DEV (work/v2-generation-scale; DESIGN.md WP3.1 eighth increment (0ea83af2); DESIGN.md WP3.1 seventh increment (fef92189) for $u\bar{u} \to ggggg$): generation in one thread (LoopSlice.generate, then the plan export; peaks of each stage), then binary64 virtuals with native per-point inputs and the production counterterm route at the fixture points, against the references committed on that branch (tests/fixtures/loop/required; fixture tolerance 2×10^{-8}). Largest deviation over the references and points, metric of Table 1; the per-reference values are in data/measured.json. For $gg \to ggggg$ the references are NJet (double-double and two binary64 runs) and the Catani operator for the poles, top-free with $N_f = 5$; its estimate column is the rotation plus reduction-direction check per point.

Process	References	Born	Finite	$1/\varepsilon$	$1/\varepsilon^2$	Estimate E	Generation + export
$u\bar{u} \to e^-e^+d\bar{d}g$	ML, Rc, OL · 3 pts	1.9×10^{-14}	8.7×10^{-13}	2.3×10^{-13}	1.0×10^{-13}	—	36.1 s 0.4 GiB
$u\bar{u} \to e^-e^+ggg$	ML, Rc, OL · 3 pts	2.3×10^{-14}	8.5×10^{-13}	4.6×10^{-13}	2.0×10^{-13}	—	36.7 s 0.5 GiB
$gg \to t\bar{t}e^-e^+g$ (QCD)	ML, Rc, OL · 3 pts	1.4×10^{-14}	5.7×10^{-9}	3.7×10^{-12}	5.3×10^{-12}	—	43.7 s 0.6 GiB
$gg \to t\bar{t}gg$	ML, Rc, OL · 3 pts	2.1×10^{-14}	1.1×10^{-11}	1.0×10^{-10}	1.6×10^{-11}	—	38.5 s 0.7 GiB
$gg \to gggg$	Rc, OL · 2 pts	4.7×10^{-14}	4.5×10^{-11}	1.2×10^{-11}	5.2×10^{-12}	—	151 s 2.2 GiB
$u\bar{u} \to ggggg$	ML, Rc · 3 pts	1.7×10^{-13}	2.9×10^{-10}	6.1×10^{-10}	4.4×10^{-11}	—	517 s 6.1 GiB
$gg \to ggggg$	NJet DD, NJet R1, NJet R2, Catani · 3 pts	8.1×10^{-13}	3.8×10^{-9}	2.5×10^{-9}	5.3×10^{-12}	2.0×10^{-8}, 6.0×10^{-8}, 1.4×10^{-10}	54.9 min 31.9 GiB

The $gg \to t\bar{t}e^-e^+g$ (QCD) finite part deviates by 5.7×10^{-9} at one of its three points and by at most 3.7×10^{-12} at the others: the point's known binary64 conditioning (earlier record, DESIGN WP3.1 fourth increment: $\lesssim 3.6\times10^{-12}$ at two points; third point 5.7×10^{-9} in binary64 (estimate 1.7×10^{-9}), MadLoop to 1.9×10^{-14} in DoubleFloat). The electroweak target is not started (WP3.2). The $u\bar{u} \to ggggg$ row comes from DESIGN.md WP3.1 seventh increment (fef92189), the newest record of that process: the eighth increment's re-run on the merged head does not include it. It uses the same production counterterm route; its check is in Table 14.

4 Seven gluons: NJet and pyAmpliCol

4.1 NJet: Table 5 of arXiv:1209.0100, re-measured on this host

NJet is the independent reference for $gg \to ggggg$, where MadLoop and Recola cannot generate under the 45-minute rule and OpenLoops has no library. It is massless five-flavour QCD, and it is not one of the gate's matched providers.

Table 13: **NJet, s per point**, double precision, two evaluations each (NJet's scaling test). Paper: Xeon E3-1240 at 3.3 GHz, NJet 1.0. This host: one pinned core, idle at launch, of the shared host; 10 frozen points per channel, mean (spread ≤ 2 %); the multiplier is time ÷ the paper's. De-symmetrised sums equal the standard sums only after phase-space integration; they are timing figures, not point values.

Channel	Table 5	NJet 3.1.2		NJet 2.1.1		Table 5	NJet 3.1.2		NJet 2.1.1	
	standard full-colour sum					de-symmetrised sum				
$gg \to gg$	0.03	**0.00295**	×0.1	**0.00291**	×0.1	0.03	*no class*		*no class*	
$gg \to ggg$	0.22	*defective channel*		**0.141**	×0.6	0.07	*defective channel*		**0.0452**	×0.6
$gg \to gggg$	6.19	**4.59**	×0.7	**4.58**	×0.7	0.57	**0.39**	×0.7	**0.38**	×0.7
$gg \to ggggg$	171.31	**142**	×0.8	**246.7**	×1.4	3.07	**2.41**	×0.8	**4.19**	×1.4

At seven gluons one NJet 3.1.2 evaluation without the scaling test takes 71.2 s (standard) and 1.22 s (de-symmetrised), at 69 MB; double-double costs 1 990 s per point. NJet 3.1.2's numerical five-gluon channel is 9/2 too large (NJet 2.1.1 is used there), and at the unrotated seven-gluon points NJet keeps only 4–6 digits in double precision (NaN at 3 of 10), so the reference values are taken at exactly rotated points.

4.2 pyAmpliCol at seven gluons DEV

On the WP3.1 branch (Table 12) the full $gg \to ggggg$ virtual, with the counterterms of the recursion and native per-point inputs, top-free with $N_f = 5$, agrees with NJet's double-double values to 3.8×10^{-9} in the finite part (per point: 3.8×10^{-9}, 3.1×10^{-9}, 4.3×10^{-11} over all references) and to 5.3×10^{-12} in the double pole; each point's rotation plus reduction-direction estimate bounds its difference. Generation took 46.1 min at 15.8 GiB and the plan export 8.8 min at 31.9 GiB (peak RSS of each stage; 160 MB of plan JSON; the whole validation run, point evaluations included, peaked at 42.2 GiB), inside addendum H's budget (hours, below 150 GB). DESIGN WP3.1 (eighth increment) attributes the cost: 325 s before the counterterm recursion, 1,122 s with its counterterm program, 2,766 s once poles-uvir's always-generated companion merged (its seed table builds generic seeds; making it lean is open). A point with its rotation and reduction-direction checks took 84.3, 41.2, 39.1 s: single-point development evaluations, unpinned, with up to eight threads (the workspace sets RAYON_NUM_THREADS=8), so not a batched one-thread time and not comparable with NJet's 71.2 s per evaluation on one pinned core. No batched seven-gluon timing exists yet; Table 1 therefore has none.

4.3 Validation without a matched provider

Table 14: **Checks of the seven-leg QCD results**, DEV on `work/v2-generation-scale`, except the Ward row (WP2.5 on `loop`); binary64, all three Laurent coefficients, deviation metric of Table 1. NJet values: fixture `gg_ggggg_njet.json` (NJet 3.1.2 double-double at exactly rotated points, scaling-test estimates $\lesssim 1.3\times10^{-24}$).

Check	Result	Status
$gg \to ggggg$ full virtual against NJet (top-free, $N_f = 5$), 3 points (DESIGN.md WP3.1 eighth increment (0ea83af2))	Born $\le 7.5\times10^{-13}$, finite $\le 3.8\times10^{-9}$, single pole $\le 2.4\times10^{-9}$, double pole $\le 5.3\times10^{-12}$; Catani poles: single $\le 2.4\times10^{-9}$, double $\le 5.3\times10^{-12}$; estimates 2.0×10^{-8}, 6.0×10^{-8}, 1.4×10^{-10}	passed (tolerance 2×10^{-8})
Top-free full virtual against NJet at lower multiplicity	$gg \to gg$: $\le 2.1\times10^{-14}$ at 10 points, full virtual; $gg \to ggg$: $\le 2.3\times10^{-13}$ at the stable points; the flagged point 3×10^{-14} in DoubleFloat; $gg \to gggg$: 1.0×10^{-11} at point 0; point 1 1.6×10^{-7} in binary64 (estimate 2.7×10^{-6}), 4.7×10^{-15} in DoubleFloat (`test_loop_v2_topfree.py`)	passed
$u\bar{u} \to ggggg$ full virtual against MadLoop and Recola, 3 points (DESIGN.md WP3.1 seventh increment (fef92189))	MadLoop $\le 2.6\times10^{-12}$, Recola $\le 6.1\times10^{-10}$ (finite $\le 2.9\times10^{-10}$); the providers agree to 6.2×10^{-10} (reference merge)	passed (tolerance 2×10^{-8})
Ward identity, one gluon's $\varepsilon \to p$, at seven gluons	asymmetric Ward mode of WP2.5 (loop side only, bare + UV + R_2); passes on $gg \to gg$ and $u\bar{u} \to ggg$ (`test_loop_v2_observables.py`)	**not yet run at seven gluons (addendum H)**

5 Capabilities at this milestone

Table 15: **Capability matrix** of the v2 engine on `loop` at b4d76cc3, with branch work marked.

Feature	Status	Boundary
Recursive engine	supported	Trees, families and loop numerators from recursions; no loop diagram is enumerated (FeynKit graphs are test oracles only). Counterterms: see below.
Full, NLC and LC colour	supported	Exact seed tables in N_c; LC/NLC by generated classes and truncated cells. LC is cheaper in the loop stage but not yet end to end at large batches (its native trees cost more, Table 6), nor to generate at seven legs.
Colour and spin correlations	supported	Directed colour histories at LC/NLC/FC, spin sources, resolved helicities, flavour aliases, Ward sources (WP2.5). Seven-leg directed LC/NLC tables need the Rust seed tables (WP3.1).
Per-order partials	supported	Every squared coupling order at tree level and at one loop, mixed Born orders, per-correction loop orders.
Poles at runtime	supported	`poles = "all" | "none" | "uv" | "ir" | "split"` (WP2.10, on `loop` since 24 October): finite-only mode with the finite part bitwise unchanged and the pole work skipped; `"uv"` and `"ir"` with the user's semantics of 23 October. The UV residue comes from an interim local module until the NinjaRS re-pin; a `"uv"` estimate needs a rotation check.
Stability estimates	supported	Rotation checks (any number, 0 = estimate absent), reduction direction, exact rescaling, in any combination; no automatic escalation (WP2.3).
Precision carriers	supported	Binary64, DoubleFloat, arbitrary precision and exact rationals in the native executor; DD/AP point inputs are formed in Python Decimal until WP3.1.
Native executor, schedules	supported	Rust, 16 lanes (also for a one-point call, a G1 lever), threads; G1 decision on the schedule-dependent stages (Table 9, provisional until compiled trees): M for pure-gluon LC from $n = 6$, B for every other class at $n \le 6$; F as oracle. The NinjaRS reduction runs serially. Development report: stage times, modelled kernel work, master-integral time.
Plan v4 artifacts and API	partial	Published and loaded through Python, TOML and CLI; relocation to another path tested on the same host and CPU. Loading regenerates the process (the `LoopSlice`) in Python and checks the plan digest, which addendum G forbids for prepared artifacts; WP3.1 removes it. The Rust/C/C++/Fortran SDK refuses v4 until WP3.1.
Counterterms (model UV, R_2)	supported	Plain tree vertices of the recursion inserted exactly once (WP2.2, a18b7599; addendum J, b4d76cc3), admitted by coupling orders and the public `particle_veto`; no witness table or loop matching in production. An unlowerable counterterm vertex that no tree of the request reaches is inert and recorded in the artifact; a reachable one is refused. Site closure is the test oracle. Per point they are formed in Python on `loop`, natively on the WP3.1 branch.
Direct R_2	deferred	A bonus by the user's rule; bounded controls only.
Electroweak, complex mass	open	Tree EW lanes fixed, the Dirac–Weyl class refused; loop EW, Goldstones and CMS are WP3.2.
Required set	in progress	On the WP3.1 branch seven of eight generate and agree with their references in all coefficients with native inputs (Table 12), $gg \to ggggg$ against NJet's double-double values; the EW target is not started. Not on `loop` yet; no batched $2 \to 5$ timing.
Performance gates	in progress	G1 measured (Section 2): the integrated slice fails every $2 \to 3$ and $2 \to 4$ gate it generates (Python per-point inputs) and does not generate V+jets (WP3.5, WP3.2). With the WP3.1 native inputs at $B = 128$: 8 of 9 pass ($u\bar{u} \to u\bar{u}g$ by user decision, 2026-10-24), $u\bar{u} \to d\bar{d}g$ fails; $B = 1$ needs lane adaptation (Table 3). Missing: provider protocol-mode columns, LC for V+jets, seven legs (`measure-g1-7leg`).
$gg \to 6g$	deferred	No $2 \to 6$ work until the $2 \to 5$ plan is complete (user, 23 October).

6 Architecture of the v2 engine

6.1 The computed quantity

For Born order a, loop order b and squared coupling order $s = a + b$,

$$V_{s,k} = \mathcal{N} \sum_{a+b=s} 2\,\mathrm{Re}\big[B_a^\dagger G_{BL} L_{b,k}\big], \quad k \in \{0,-1,-2\}, \quad V = C_\varepsilon \left(V_0 + \frac{V_{-1}}{\varepsilon} + \frac{V_{-2}}{\varepsilon^2}\right), \quad C_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)},$$

where G_{BL} is the colour metric between Born and loop colour structures and $\mathcal{N}$ holds the initial-state averages and identical-particle factors (MadLoop/BLHA convention; couplings included, nothing divided by the Born). Directed colour and spin correlations replace $2\,\mathrm{Re}(\ldots)$ by the two cross terms $B_A^\dagger G L_B + L_A^\dagger G B_B$.

6.2 Pipeline

For every denominator family f, sector s and observable column o, v2 builds the four-dimensional numerator $N_{f,s,o}(q) = \sum_\alpha c_\alpha q^\alpha$ that NinjaRS reduces: Born-weighted and helicity-summed, produced by three recursions that share all work (Figure 2).

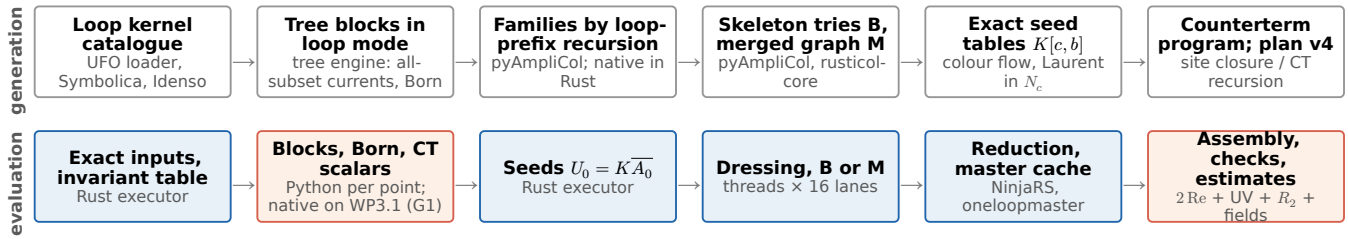


Figure 2: **v2 pipeline.** Upper lane: generation, once per process, into a plan v4 artifact. Lower lane: evaluation per batch of points. Blue: native batched numerics in the Rust executor; orange: still formed in Python per point (native lanes on the WP3.1 branch); white: generation. The owning code or dependency is named under each step.

Trees. The tree engine's colour-ordered Berends–Giele recursion computes, once per point, every off-shell current over every leg subset a loop vertex can absorb (the *blocks*), the Born colour components A_0 and, with WP2.2, every tree with one counterterm vertex. A loop-aware reachability test keeps exactly the currents some family position or root needs. Blocks with transverse sources use the exact transverse Yang–Mills kernels; loop segments, counterterm currents and Ward sources use the full ones.

Families and skeletons. A Recola-type recursion over loop prefixes (ordered leg sets, propagator species and masses) yields the denominator families, keyed by the canonical cyclic sequence of (leg mask, mass) and equal to Recola's tensor-integral identities. Per family it yields the colour-ordered *skeletons*: rings of blocks, each block summing all trees on its leg subset (Figure 3). Skeletons are never stored as a list; they are the leaves of the schedule tries. Only certified structural zeros are dropped (scaleless rings, external wave-function corrections, VEV tadpoles, colour zeros); nothing is restricted to 1PI.

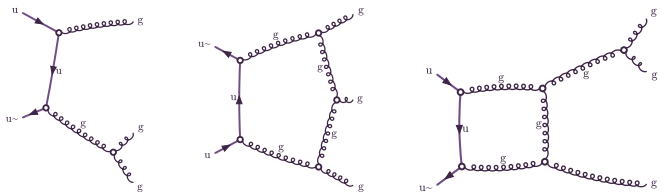


Figure 3: Three graphs of $u\bar{u} \to ggg$ drawn by FeynKit's renderer from its own generation (an oracle, not a production input). In v2 the Born (left) is a closure of blocks; the pentagon (middle) is a ring of five single-leg blocks; the box with a tree attachment (right) is a ring of four blocks, one of them the two-leg current, computed once per point and shared by every skeleton that contains that leg subset.

Loop numerators and schedules. An open-loop recursion dresses skeletons one segment at a time with polynomial-valued objects, $O' = O \cdot Y + \mathrm{raise}\big(O \cdot Z_\mu, q^\mu\big)$, in the NinjaRS monomial layout ($\binom{r+4}{4}$ coefficients at rank r, no rank cap). The seed is the colour-summed Born interference, and each block's helicities are summed as it is attached. One generator feeds three schedules (Table 16).

Table 16: **Three schedules on one generator** (DESIGN §2.2, §5.6).

	F: forward	B: suffix trie (default)	M: merged, primitive-seeded
Born weighting	at closure	at the leaves (one seed per skeleton class)	at the cut, one seed per colour ordering
Shared	prefixes across families	all prefixes with equal family and remainder	prefixes across families and compositions
Colour certificate	none needed	none needed	certified classes: adjoint rings, traces, quark rings
Role	oracle	default; generic fallback	pure-gluon LC from $n = 6$ (G1); planned default from $n \geq 8$

Colour. Colour lives only in exact seed tables $K[c, b]$, Laurent polynomials in N_c between seed classes c (the colour vector obtained by closing the cut of a skeleton's colour flow) and Born basis elements b; per point the seeds are $U_0 = \sum_b K[c, b]\overline{A_0[b]}$, with a symmetric-group FFT where certified. LC and NLC differ from full colour only in which classes are generated and how cells are truncated: at LC each colour ordering is seeded by its own Born amplitude and skeletons below the leading N_c degree are never generated. Certificates (adjoint rings, fundamental traces, quark rings) are proved at loop level, never inherited from the tree level.

Reduction and masters. Families close directly into NinjaRS `TensorNumerator` lanes: one reduction per (family, sector, column), families with equal (propagators, rank) batched as lanes of one call, scalar masters from oneloopmaster cached per event. A second reduction in a relabelled frame reuses the numerators and the cached masters as the cheap stability check.

Counterterms. The model's UV and R_2 vertices, two-point ones included, are ordinary vertices of the tree recursion, and each contribution contains exactly one of them; external field factors are separate sources. Admission is the requested coupling orders, the exactly-once constraint and a public `particle_veto`: a coupling term whose loop particles include a vetoed particle is dropped, and a vetoed particle is removed from every tree vertex, loop line and propagator of the generation, so loops and counterterms stay consistent. Nothing is matched to the loop (user, 23 October; addendum J). On `loop` the counterterms come from the WP2.2 recursion (one counterterm order in the tree recursion, the default route, whose $gg \to ggggg$ program leaves no rule to site closure) with this admission (`ct-simplify`, b4d76cc3): no witness table and no loop matching. A counterterm vertex that the model cannot lower and that no tree of the request reaches (the tree engine's own reachability, with the program's order limits and `particle_veto`) is inert and recorded in the artifact; a reachable one is refused. `particle_veto = [t]` replaces the internal top-free switch of Table 14; site closure and the witness machinery remain only as test oracles.

Precision, stability and assembly. Every kernel is generic over binary64, DoubleFloat, arbitrary-precision floats and exact rationals. A failed lane fails its point and never branches a kernel. The assembly takes $2\,\mathrm{Re}$ of $i/(16\pi^2)$ times the reduced family sum, plus local UV, R_2 and field terms, per sector and column, with optional rotation, reduction-direction and rescaling estimates and the finite-only mode. The public API selects `loop_engine = "v2"` from TOML, CLI or Python and stores plan v4 as a relocatable PACBIN member (zlib JSON of exact kernel term lists, no compiled code, digest checked at load).

Table 17: **Asymptotic classes per event** for n gluons, full colour (DESIGN §1.4): $F_n \approx (n-1)!\left(\frac{1}{\ln 2}\right)^n/2$ families, 2^n helicities. The required set ($n \leq 7$) is far below the crossover of schedule B with the unitarity class ($n \approx 30$–50); schedule M is planned as the default from $n \geq 8$.

Stage or code	Class
Tree blocks	$\Theta(n \cdot n!)$ currents; $\Theta(n! \cdot 2^n)$ current-helicity rows
Dressing, schedule B	$\Theta((n-1)! \cdot 4^n)$ seed injection; trie dressing dominates for $n \leq 8$
Dressing, schedule M	$\Theta(n! \cdot 2^n \cdot n^4)$
NinjaRS	$\Theta(F_n)$ reductions, $\Theta(F_n \cdot n^8)$ work, $\Theta(4^n)$ distinct masters after caching
MadLoop (per diagram, per helicity)	$\Theta(D_n \cdot 2^n \cdot n^4)$, D_n super-exponential ($gg \to gggg$: 26 975 loop diagrams)
Unitarity with exact helicity sums (NJet)	$\Theta((n-1)! \cdot 2^n \cdot n^{6\ldots9})$

Table 18: **Ownership of the mathematics.** pyAmpliCol reuses its dependencies for algebra, tensors, graphs and integrals and owns the recursions and the assembly.

Component	Responsibility
ufo_model_loader	Lossless UFO loop-model import: Laurent-valued parameters and couplings, UV/R_2 categories, loop groups
Symbolica	Expressions, coefficient extraction in q, evaluators; exact rationals and arbitrary precision
Spenso, Idenso (Community)	Lorentz and colour tensors behind the kernel catalogue and the colour metrics
FeynKit (Community)	Test oracles: graph and family counts, overall factors, drawings (Figure 3)
NinjaRS, oneloopmaster	Integrand reduction with Laurent expansion; scalar masters in all carriers
pyAmpliCol tree engine	Blocks, Born, coupling orders and colour flow (Python lane; Rust recurrence)
pyAmpliCol loop v2	Families, skeletons, schedules, seed tables, counterterm program, plan v4, native executor, API

7 Sources and reproducibility

- **Build:** `typst compile benchmarks.typ benchmarks.pdf` in `docs/performance_reports/loop/` (Typst 0.15).
- **Status, in-flight work, decisions:** `data/milestone.json`, from the `Integrate …` merges of b4d76cc3 since f132e53c; `workspace/reports/engine-v2/slice-status.md` (integration blocks, DECISIONS 2026-10-23/24; the user's gate decision of 2026-10-24, 16:05 UTC, is recorded in `data/milestone.json` `gate_decisions`); `HANDOFF_24_10_2026.md`; MASTER_PLAN 2026-10-23/24 entries; REQUIREMENTS_ADDENDUM A, D, F, G, H, I, J; DESIGN §5.7 (G1), §6.1, §6.2, §12.4; the 24 October pushes (59badc7a and b4d76cc3, gates in slice-status.md).
- **Table 1, Figure 1:** `data/summary.json`, written by `scripts/build_summary.py` from `data/g1.json`, `data/measured.json`, `data/milestone.json` and `tools/loop_benchmark/processes.json`; every entry names its source (README).
- **Table 11:** HANDOFF §0.1 (addendum F); the provider generation records of Table 1 (`data/summary.json`): `data/g1.json` for the G1 rows and the reference runs' records (`workspace/reference/runs/required/<process>/`, collected into `data/measured.json`) for the $2 \to 5$ rows; MadLoop's production-build times of the $2 \to 5$ rows: the same runs' `madloop-production-result.json`.
  Table 12, Section 3, Section 4: the native validations on `work/v2-generation-scale` (DESIGN.md WP3.1 eighth increment (0ea83af2); DESIGN.md WP3.1 seventh increment (fef92189) for $u\bar{u} \to ggggg$), records `workspace/tmp/generation-scale/gint/val1/`, `val2/` and `i7/val/`, collected into `data/measured.json`; the earlier $gg \to t\bar{t}e^-e^+g$ record: DESIGN WP3.1 fourth increment. **Conventions:** MadLoop standalone, Recola, OpenLoops 2.
- **Section 2 (Table 2 to Table 10):** `data/g1.json` and the per-point values `data/g1-values.json`, written by `python -m tools.loop_benchmark gate` (README and docstrings of `tools/loop_benchmark`); streams `tools/loop_benchmark/streams/`; run records archived in `workspace/reports/engine-v2/measure-g1/records/` (first campaign `runs/`, fix round `runs2/`), MadLoop exports of the G1-only processes in `workspace/reference/runs/g1/`; recalibration `tools/loop_v2_costmodel.py --g1 data/g1.json`. Builds: `loop-native-measure-g1-033edf15` (integrated, this branch at `1c971289`) and `loop-native-measure-g1-genscale-39824f9b` (`work/v2-generation-scale` `889af02c` plus the timers; scratch commit `39824f9b`, kept as tag `g1-native-inputs-39824f9b`), release with one codegen unit and thin LTO; the native-input harness is this branch's `tools/loop_benchmark`. Schedule M at seven legs: DESIGN §5.6.
- **Table 13** and NJet prose: `workspace/reports/engine-v2/njet-benchmark.md` §2, §5, §6 (raw `workspace/reference/runs/njet/timing/`); NJet, arXiv:1209.0100 Table 5.
- **Table 14:** `data/measured.json` (`scripts/collect_milestone.py <workspace>`); top-free values at lower multiplicity: DESIGN WP3.1 and `test_loop_v2_topfree.py`; the seven-gluon generation history: DESIGN WP3.1 eighth increment. Memory in GiB = 2^{30} bytes.
- **Table 15, Section 6:** `docs/development/loop_v2/DESIGN.md` §1–§3, §5.5, §5.6, §6, §10, §12, REQUIREMENTS_ADDENDUM J (counterterm rule) and the integration records above.
- **History:** the legacy per-diagram engine's snapshot of 2026-10-18 (`data/measurements.json`, `provider-cold.json`, `legacy-labels.json`, `colour-measurements.json`) stays for the DESIGN citations and is no longer shown; its table is in the earlier report revisions (README).